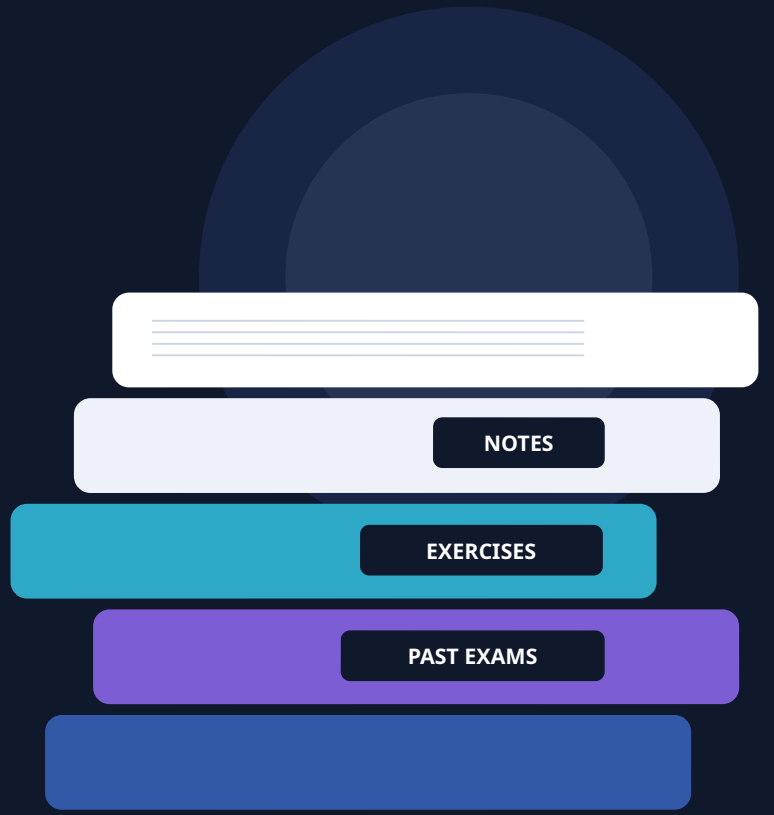
UNIVERSIDADE DE AVEIRO / PROJETO EM ENGENHARIA INFORMÁTICA

ApontaUA

Collaborative Academic Document Management System

MILESTONE 2 · LIFECYCLE ARCHITECTURE

Centralize. Discover. Contribute. Validate.



PEI Group 17

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Today: from evidence to an implementable MVP

A short walkthrough of what we learned and how it shapes the design.

01 **Problem & evidence** — What students struggle with today

02 **State of the art** — What existing platforms already do

03 **Requirements & use cases** — What ApontaUA must support

04 **Architecture & domain model** — How we plan to build it

05 **Mockups, risks & MVP** — What comes first, what can wait

THE THROUGHLINE

Find it.
Trust it.
Reuse it.

A visible lifecycle
for shared materials.

The problem is not a lack of files

Student responses show fragmented discovery; teacher responses are more mixed.

01

Scattered materials

Moodle, shared drives, messaging apps, websites, classmates and personal notes.

02

Repeated effort

Different cohorts recreate summaries, exercises and exam collections instead of reusing existing work.

03

Mixed teacher feedback

3/6 reported no major problem; 2/6 want student feedback; 1/6 cited outdated materials.

WHERE MATERIALS LIVE TODAY

MOODLE | GOOGLE DRIVE | WHATSAPP | ONE DRIVE | PEERS | UA WEBSITES

Opportunity: make academic context, freshness and review status visible — not just store another copy.

Sources: student questionnaire (n = 52) and teacher questionnaire (n = 6), October 2026.

State of the art: existing solutions solve different parts

The comparison is about purpose and workflow, not a claim that one tool is universally better.

LOCAL REPOSITORY

Resumos de Informática

STRENGTH

Year / subject organization

LIMIT / GAP

Team observed outdated items, duplicates and legacy UCs.

UA COMMUNITY

NEI Apontamentos

STRENGTH

Filters for academic year, curricular year, subject, author, professor and category

LIMIT / GAP

Visible academic-year options span 2013/14–2022/23. Contributions are invited by email; per-material version history and review status are not evident on the public page.

STUDENT SHARING

Studocu

STRENGTH

Search filters and upload screening

LIMIT / GAP

Uniqueness checks do not prove academic correctness.

COURSE LMS

Moodle

STRENGTH

Resources in teacher-managed course spaces

LIMIT / GAP

Not primarily a shared cross-cohort repository.

CURATED OER

OpenStax

STRENGTH

Expert-authored, peer-reviewed textbooks

LIMIT / GAP

Not focused on local student notes and contributions.

Sources: MS1 team review; NEI [4] (public page reviewed 12 Oct 2026), Studocu [5–6], Moodle [7], OpenStax [8].

Where ApontaUA intends to differentiate

"Not evident" means the reviewed material did not establish a feature — not proof that it is absent.

SOLUTION	ACADEMIC STRUCTURE	CONTRIBUTION	DUPLICATE HANDLING	HISTORY / VALIDATION	MAIN FOCUS
Resumos de Informática	Year / subject	Materials available	Duplicates observed	Not evident in MS1 review	Informatics
NEI Apontamentos	Year, UC year, subject author, professor, category	Submit via email	Not evident on reviewed page	Per-item history / review status not evident	Informatics student community
Studocu	Institution, course, category, year	Direct uploads	Uniqueness screening	Academic validation not established	Multi-institution
Moodle	Course spaces	Teacher-managed resources	Not a core feature reviewed	Course management; not shared lifecycle	LMS
OpenStax	Textbook catalogue	Expert-authored	Not primary function	Expert peer review & updates	Curated OER
ApontaUA (planned)	**Course → UC → topic → material**	**Students + teachers**	**Hash + similarity warning**	**Version history + review status**	**University of Aveiro**

NEI already offers rich filters; ApontaUA aims for UA-wide coverage, in-app contribution, UC year/semester, version history and explicit review status.

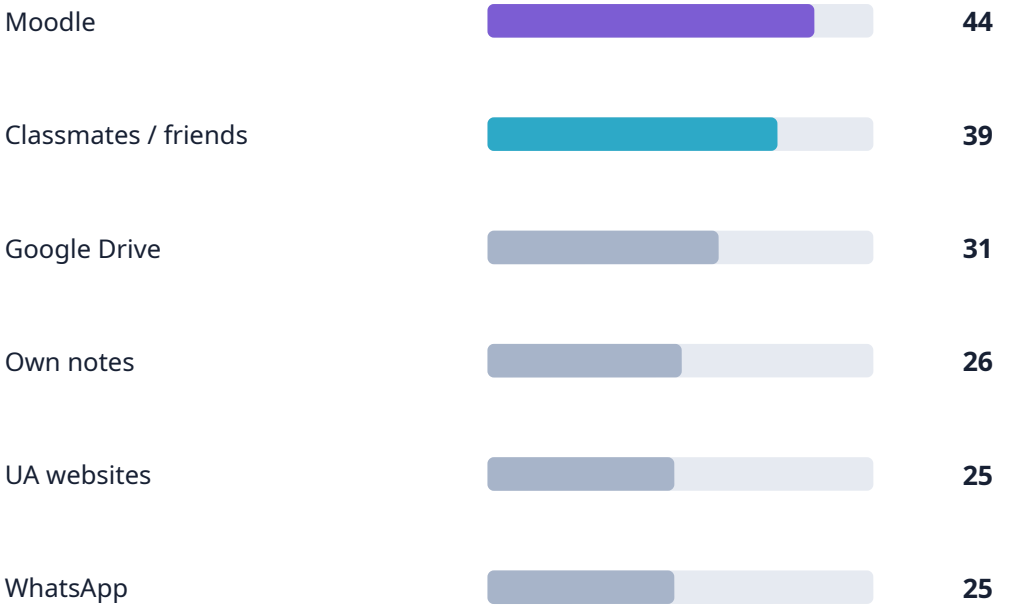
Comparison scoped to team MS1 observations and public documentation [4–9].

Student survey: discovery is fragmented

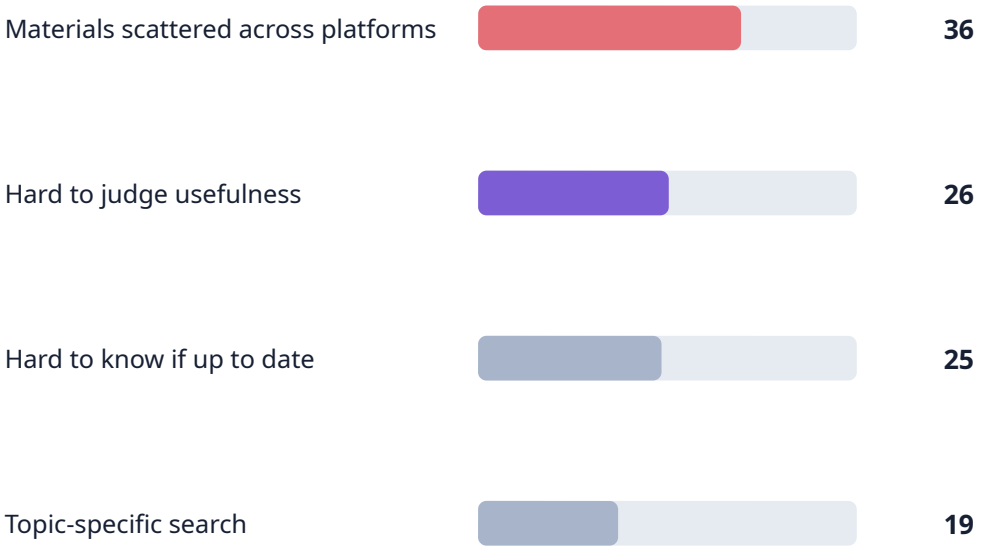
52 student responses · exploratory evidence, not a representative survey of the entire UA.

WHERE STUDENTS LOOK

MULTIPLE ANSWERS ALLOWED



MAIN DIFFICULTIES



Profile caveat: 46/52 respondents were in year 2; 47/52 were in Informatics-related variants.

Source: ApontaUA student requirements questionnaire (n = 52). Counts are multi-select where applicable.

Student demand is strong; teacher views are mixed

Teacher responses support visible validation, but views on a new aggregator differ.

98%

51 / 52

Search past exams

The most requested material type.

87%

45 / 52

Review status matters

Students: 45/52 rated it 4–5; teachers: 5/6 rated validation labels 4–5.

92%

48 / 52

Duplicate warnings?

Students: 48/52 said yes/maybe; teachers: 2 yes, 4 maybe.

3 / 6

rated it 5 / 5

Aggregator usefulness

Views were mixed: 2/6 rated it 3/5; 1/6 rated it 1/5.

TEACHER QUESTIONNAIRE — EXCERPTS (ORIGINAL PORTUGUESE)

"Considera útil existir uma plataforma que organize materiais por curso, unidade curricular e tópico?"
Result: 3/6 rated 5/5, 2/6 rated 3/5, 1/6 rated 1/5.

"Considera importante que os materiais submetidos pelos estudantes possam ser identificados como 'revistos/validados' ou 'não revistos'?"
Result: 5/6 rated this distinction 4/5 or 5/5.

Design implication: keep validation visible; duplicate detection should warn, not auto-delete.

Sources: "ApontaUA — Questionário de Requisitos para Docentes" + response export (n=6); student questionnaire (n=52). Exploratory samples.

From mixed evidence to focused product goals

Student feedback points to fragmented discovery; teacher views on a new aggregator are mixed.

01

Flexible discovery

Search, filters and tags; support UC/topic browsing without forcing one rigid hierarchy.

02

Enable collaboration

Enable contributions and peer feedback, keep versions traceable and preserve accessible variants.

03

Make trust visible

Show review status, authorship, source and licence; similar items prompt human inspection.

DESIGN PRINCIPLES

Complement, don't duplicate

Teacher views on another aggregator were mixed; show value beyond Moodle and LLMs.

Peer feedback ≠ academic approval

Lecturer defines UC-specific criteria; peer feedback informs the final decision.

Warn, don’t over-block

All 6 often see similar items; 2/6 said yes to detection, 4/6 maybe. Never auto-delete.

Sources: student questionnaire (n=52), teacher questionnaire (n=6), and project requirements; exploratory samples.

Use cases: three roles, distinct permissions

Proposed policy: the responsible UC lecturer defines review criteria and gives final academic approval; peer feedback informs the decision.

S STUDENT

- Browse courses / UCs / topics
- Search and view materials
- Publish and edit own material
- Review versions, comment and rate

T TEACHER

- Define UC criteria: scientific rigour, syllabus fit, currency and completeness
- Require author, source/DOI, references, update date and licence
- Set accessibility expectations; review peer feedback and request corrections
- Give final academic approval or revoke validation

A ADMINISTRATOR

- Manage users and roles
- Manage course / UC structure
- Moderate reports and content
- Review audit events / platform rules

PROPOSED MATERIAL LIFECYCLE

Submit → **Check format / duplicates** → **Not reviewed** → **Peer review + UC criteria** → **UC lecturer approves**

Proposed governance: lecturer defines UC-level criteria; peers advise; lecturer records the final academic decision.

Functional baseline: the MVP must support a complete material lifecycle

These are the high-value capabilities; detailed FR catalogue remains the source for implementation and testing.

01 **Browse & search**

Navigate course → UC → topic; search titles and extracted text.

02 **Submit materials**

Upload material, declare source/author/licence and preserve the original.

03 **Manage versions**

Editing creates a new version; history remains accessible.

04 **Detect likely duplicates**

Exact hash first; similarity checks show candidates, not automatic bans.

05 **Review & validate**

Record reviewer and outcome; the responsible UC lecturer makes the final decision.

06 **Community & governance**

Comments, ratings, reports, rights/licence declarations and audit trail.

Source: Functional Requirements draft; student and teacher questionnaires.

Source: ApontaUA Functional Requirements draft; student questionnaire.

Architecture decisions: simple now, extensible later

One modular backend deployment. Add infrastructure only when there is a concrete need.

FRONTEND	**React + TypeScript**	TanStack Router · TanStack Query
BACKEND	**Python + FastAPI**	REST API · OpenAPI · modular monolith
DATA	**PostgreSQL**	Relational source of truth · full-text search first
AUTH	**Keycloak (MVP)**	UA institutional login planned for second semester
FILES	**Persistent file volume**	Original documents separate from metadata
PROCESSING	**Python worker + Celery / Redis**	Conditional: include if long-running OCR/jobs are in MVP

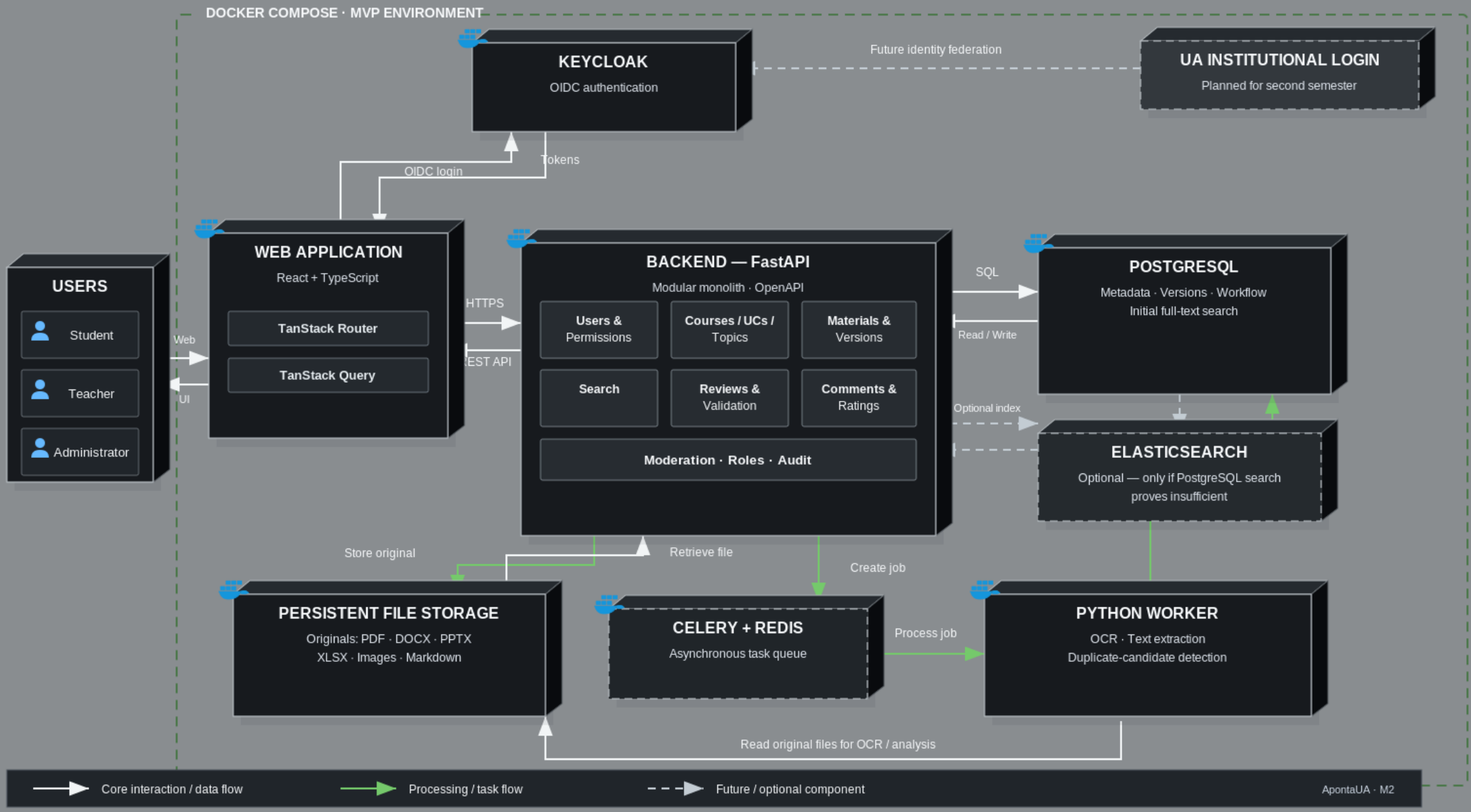
Elasticsearch: deferred until PostgreSQL proves insufficient.

Microservices: not justified for the MVP.

Technology decisions agreed by the group; processing scope remains conditional.

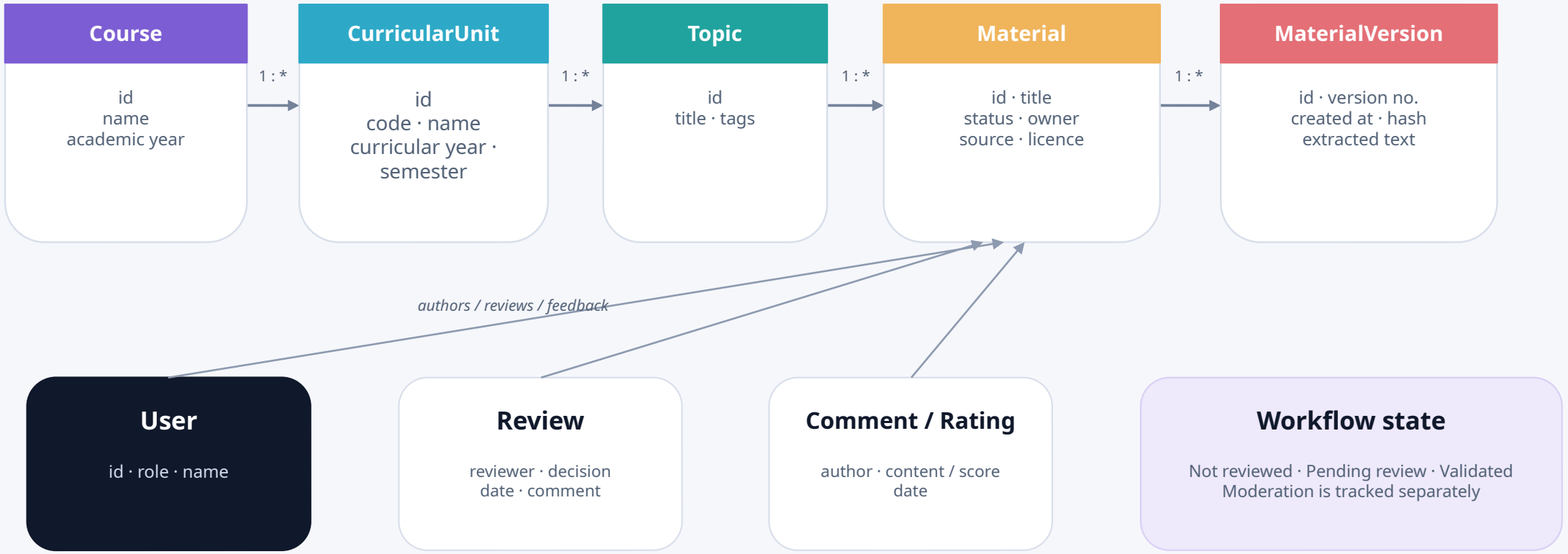
APONTAUA — SYSTEM ARCHITECTURE

Modular Monolith · REST API · Docker Compose



Domain model: the academic hierarchy anchors everything

Initial conceptual model — entity attributes and cardinalities will be refined during implementation.



Proposed model includes provenance/licence metadata; accessible variants must remain linked.

Proposed UI direction: one library, clear status, traceable changes

Concept wireframes for discussion — not final visual design or implemented screens.

APONT AUA / LIBRARY

Search materials, topics, courses...

Filters

- Course: Informatics
- Unit: Testing & Quality
- UC year: 3rd · Semester: 1st
- Topic: Test design

RECENT MATERIALS

Past exam - 2025	Validated
Testing summary	Not reviewed

MATERIAL DETAIL

Software Testing — Summary

VALIDATED BY TEACHER

Author: student · Source link

Updated 12 Oct · Licence declared

VERSION HISTORY

v3	Current	12 Oct
v2	Updated notes	08 Oct
v1	Initial upload	02 Oct

SUBMIT MATERIAL

1. Upload file + declare rights

Drop a file here or browse

2. Context, source & licence

- Course: Informatics
- Unit: Testing & Quality
- Topic: Test design
- UC year: 3rd · Semester: 1st

Possible duplicate found

Review candidate before continuing.

Illustrative UC year/semester values; final metadata should follow the official UA curriculum.

Non-functional baseline: trust must be designed in

Quality constraints guide implementation even when exact performance targets are still open.

Security & permissions

Use role checks; uploaders confirm rights to share; provide a reporting/removal workflow.

Integrity & traceability

Preserve originals; record authorship, source/DOI when available, licence and review decision.

Usability & accessibility

Platform target: WCAG 2.2 AA (to validate). Keep accessible adaptations (e.g., dyslexia-friendly versions) as linked variants, not duplicates.

Reliability & operation

Persistent storage and backups; processing status/error handling; define availability and recovery targets before pilot.

Capacity assumption

Design scenario: ~300 registered users. Concurrent load, response-time thresholds and uptime target are to be defined and measured before the pilot.

Sources: ApontaUA NFR draft; teacher questionnaire; W3C WCAG 2.2.

Main risks and how we will reduce them

Key risks include scientific quality, unclear reuse rights, duplicate handling, accessible variants and operational scope.

RISK	IMPACT	MITIGATION
Incorrect / outdated material	HIGH	Record author and update date; the responsible UC lecturer checks scientific correctness and course alignment.
Duplicate or near-duplicate noise	MEDIUM–HIGH	Exact hash + similarity warning; never auto-delete content; preserve accessible adaptations as linked variants.
OCR or extraction errors	MEDIUM–HIGH	Keep original; surface uncertain extraction; permit correction and retry.
File or database loss	HIGH	Persistent volumes, backup plan and restore test before pilot.
Copyright / unclear licence	HIGH	Confirm uploader has rights to share; record author, source/DOI if available and licence; provide report/removal path.
Scope / operational complexity	HIGH	Modular monolith; defer Elasticsearch and optional processing infrastructure until justified.

COST DRIVERS No paid platform is assumed for the MVP. Estimate hosting, storage/backups, domain and operational time once deployment options are agreed.

Exploratory risks; validate criteria, rights workflow and permissions with the UA.

MVP scope and milestone plan

Presentation: 13 October 2026 · Official M2 architecture validation date: 20 October 2026.

MVP / THIS PHASE

- Keycloak authentication and basic roles
- Course → UC → topic navigation
- Upload / search; capture author, source and licence
- Version history, review status and lecturer sign-off
- Duplicate warnings; preserve accessible variants

FUTURE / CONDITIONAL

- UA institutional authentication
- Advanced OCR / handwritten content evaluation
- Celery + Redis if asynchronous jobs are needed
- Elasticsearch if search evidence justifies it
- Advanced recommendations / reputation features

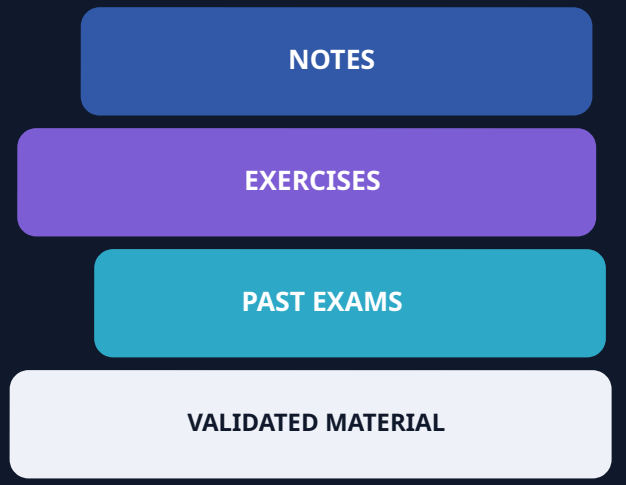
MILESTONE TIMELINE



Dates: Calendario PEI 2026/27; presentation date is earlier than the official M2 deadline.

From scattered files to a trusted academic collection.

A consistent hierarchy. A traceable document lifecycle.
Visible review status. Reuse across cohorts.



FIND
Course → topic

TRACE
Version history

TRUST
Review status

QUESTIONS & DISCUSSION

We welcome feedback on the scope, validation workflow and architecture choices.

References & sources

Sources used for the state-of-the-art comparison and the project evidence.

[1] Clements, Pawlowski & Manouselis (2015). Open educational resources repositories literature review. Computers in Human Behavior, 51, 1098–1106. DOI: 10.1016/j.chb.2015.03.026

[2] Clements & Pawlowski (2012). User-oriented quality for OER: reuse, quality and trust. Journal of Computer Assisted Learning, 28(1), 4–14. DOI: 10.1111/j.1365-2729.2011.00450.x

[3] UNESCO (2019). Recommendation on Open Educational Resources (OER). unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer

[4] NEI / AAUAv. “Apontamentos” (reviewed 12 Oct 2026). https://nei.web.ua.pt/notes

[5] Studocu Help Centre. “How do I find documents on Studocu?” help.studocu.com

[6] Studocu Help Centre. “What happens after you upload a document?” help.studocu.com

[7] MoodleDocs. Resources, Moodle 5.0 documentation. docs.moodle.org/500/en/Resources

[8] OpenStax. Higher Education — free, peer-reviewed textbooks. openstax.org/higher-education

[9] ApontaUA team. MS1 presentation, slides 5–6 (2026/27). Internal project material.

[10] ApontaUA team. “Questionário de Requisitos para Docentes” (form) and response export (n=6), October 2026; student questionnaire (n=52).

Design guidance: W3C WCAG 2.2 — w3.org/TR/WCAG22; Creative Commons licensing — creativecommons.org/faq.